

Lilian Linh Tran

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EDUCATION

San Jose State University, San Jose, CA
Master of Science in Software Engineering
Aug. 2024 – May 2026

University of California, Berkeley, Berkeley, CA
Bachelor of Arts in Computer Science & Data Science
Aug. 2019 – May 2023

- **Upsilon Pi Epsilon (UPE):** Member of the honor society for computing and information disciplines
- **Course Staff:** Structure and Interpretation of Computer Programs, Discrete Mathematics and Probability Theory
- **ICPC:** Top 30 Pacific Northwest Region, Rank 8 Northern California

RELEVANT SKILLS

Languages: Python, Java, Go, SQL, Kotlin, TypeScript

Backend/Systems: Data structures, Algorithms, Distributed systems, REST APIs, PostgreSQL, MongoDB, Linux

Testing: pytest, JUnit

Cloud & DevOps: Docker, AWS, GCP, Kubernetes, Terraform, CI/CD, Git, Agile (SDLC)

EXPERIENCE

Software Engineer

Saigon Fabrication, San Jose, CA
Nov. 2023 – Jul. 2024

- Built distributed data processing services using client-server architecture, improving system reliability and accelerating analysis of large datasets by 10%
- Designed and implemented high-throughput REST APIs and backend services using Python and Django
- Developed internal debugging and monitoring tools to diagnose issues across distributed services
- Automated validation workflows and CI/CD pipelines to streamline continuous delivery and improve deployment reliability
- Monitored and troubleshoot production workflows to maintain performance, availability, and data integrity

AI Research Assistant

The University of Texas Health Houston, Houston, TX
May. 2023 – Oct. 2023

- Designed scalable data processing pipelines with parallel computation, improving throughput and efficiency for large-scale datasets
- Developed NLP-based systems for clustering and retrieval across 69K+ medical code entities, improving data accessibility
- Applied dimensionality reduction (PCA, LSA, LDA) to optimize high-dimensional data processing
- Implemented and evaluated ML models and integrated them into backend workflows with reproducible pipelines

AI Engineer Intern

TMA Solutions, Ho Chi Minh City, Vietnam
May 2022 – Aug. 2022

- Devised data preprocessing pipelines for 2M+ data points, improving model precision by 5%
- Built GPU-enabled training environments and CI/CD workflows to support efficient training and deployment

PROJECTS

Tutor Scheduling App (*Next.js, React, TypeScript, PostgreSQL, Prisma, Supabase, NextAuth, Nodemailer*)

- Built and owned a full-stack system with backend services and APIs to support concurrent scheduling workflows
- Designed concurrency-safe database architecture to ensure consistency under concurrent access (PostgreSQL, Prisma, Auth0)

BookTable – Full-Stack Restaurant Reservation Platform (*React, TypeScript, FastAPI, PostgreSQL, Firebase, Google Maps API*)

- Developed and debugged full-stack functionality, resolving issues in search, pricing filters, and review workflows while improving application reliability and user experience.
- Integrated Firebase Cloud Storage for image uploads, Google Maps API for restaurant location services, and email notifications, while creating mock database data to support backend development and testing.

LEADERSHIP & ACTIVITIES

Senator, De Anza Student Government: Managed \$1M+ annual budget supporting campus organizations